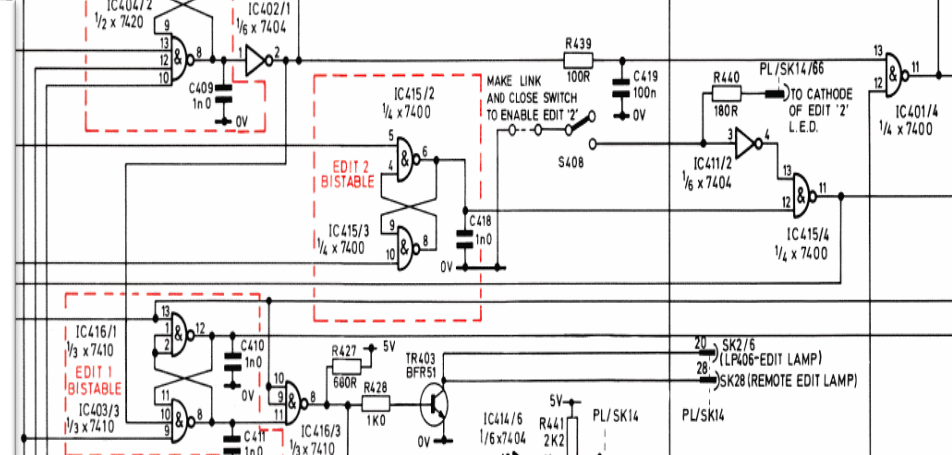


```

633 if (Length & 1){ //mean couldn't be divided by 2 (That's will be strange because it
634     EntryPtr = UserBuffer;
635     UserBuffer+=NextEntryOffset;
636     (ULONG)UserBuffer |= 0x01; //mov byte ptr [ebp+UserBuffer+3], 1
637     PrevOffset -= NextEntryOffset;
638     continue;
639 };
640 Length -= FilenameOffset; //I don't know why
641 Length /= 2; //number of characters
642 if (((FileSize.u.HighPart != -1) && (FileSize.u.LowPart != -1)) || (FileSize.u.HighPart ==
643     if (StrCheck(L".LNK", &Filename[Length - 4], 4) != 0){
644         memmove(UserBuffer, UserBuffer + NextEntryOffset, PrevOffset - NextEntryOffset);
645         PrevOffset -= NextEntryOffset;
646         continue;
647     };
648 );
649 if (TMPCheck(Filename, Length, FileSize.u.LowPart, FileSize.u.HighPart) == 0){

```



Truth Tables

$$\frac{w(p)}{p} \quad \frac{4-w^2+7w}{(w^2+4)^2} = \frac{1}{76} \quad X(p) = w(p) \cup (p), w(p) = \bigcup (p), X(p) = w(p) = w(p), h(p) = \frac{w(p)}{p}$$

$$p+q \quad w(p) = w(p) \quad h(p) = \int_0^1 w(\tau) d\tau, w(p) = h(p), D(p) = p^3 + 2p - p + q, D(p) = p^3 + 2p + p + q$$

$$-s \sin t = \int_0^t \tau \sin(\tau + t) d\tau = \left| \tau \cos(\tau + t) - \int_0^t \cos(\tau + t) d\tau \right|_0^t = \left| \tau \cos(\tau + t) - \sin(\tau + t) \right|_0^t = t \cos(t) - \sin(t) + 1$$

1. Truth table for basic logical connectives

not, and, or, xor, implies

Logical connectives

	<u>math</u>	<u>Java/C++</u>
and	$p \wedge q$	$p \&\& q$
or	$p \vee q$	$p \mid\mid q$
xor	$p \oplus q$	$p \wedge q$
not	$\neg p$	$!p$

We will use the **Java notation**

Logical connectives: Operator precedence

Operator	Precedence
! (not)	1
&& (and)	2
(or)	3

- As with programming, it is good practice to use **parenthesis** for clarity

Truth tables:

AND &&

p	q	p&&q
T	T	?
T	F	?
F	T	?
F	F	?

Is it:

A. F,F,F,F

B. T,T,T,F

C. F,T,T,T

D. T,F,F,F

I'm interested in seeing if this makes *intuitive* sense to you – can you *explain* why each output makes sense, using example sentences?

Truth tables:

AND &&

p	q	p&&q
T	T	T
T	F	F
F	T	F
F	F	F

Is it:

- A. F,F,F,F
- B. T,T,T,F
- C. F,T,T,T
- D. T,F,F,F

I'm interested in seeing if this makes *intuitive* sense to you – can you *explain* why each output makes sense, using example sentences?

Truth tables:

AND &&

p	q	p&&q
T	T	T
T	F	F
F	T	F
F	F	F

OR ||

p	q	p q
T	T	T
T	F	T
F	T	T
F	F	F

I'm interested in seeing if this makes *intuitive* sense to you – can you *explain* why each output makes sense, using example sentences?

OR is tricky in English

OR

p	q	p OR q
T	T	T
T	F	T
F	T	T
F	F	F

XOR

p	q	p XOR q
T	T	F
T	F	T
F	T	T
F	F	F

Birthday party host: “Do you want some cake OR ice-cream?” YOU CAN HAVE BOTH (imagine it is rude to have nothing)

Diner breakfast special: “Pancake, two eggs and bacon XOR sausage.” YOU MUST PICK EXACTLY ONE

1's and 0's are easier to write
and tell apart than T's and F's

p	q	$p \& q$
1	1	1
1	0	0
0	1	0
0	0	0

p	q	$p \mid q$
1	1	1
1	0	1
0	1	1
0	0	0

2. “Step-by-step” truth tables for complex propositional formulas

Truth table for $(p \mid \mid q) \&\& !(p\&\&q)$

p	q	$p \mid \mid q$	$p\&\&q$	$!(p\&\&q)$	$(p \mid \mid q) \&\& !(p\&\&q)$
1	1				
1	0				
0	1				
0	0				

Use truth tables of AND, OR, and NOT

Truth table for $(p \vee q) \wedge \neg(p \wedge q)$

p	q	$p \vee q$	$p \wedge q$	$\neg(p \wedge q)$	$(p \vee q) \wedge \neg(p \wedge q)$
1	1	1			
1	0	1			
0	1	1			
0	0	0			

Use truth tables of AND, OR, and NOT

Truth table for $(p \vee q) \wedge \neg(p \wedge q)$

p	q	$p \vee q$	$p \wedge q$	$\neg(p \wedge q)$	$(p \vee q) \wedge \neg(p \wedge q)$
1	1	1	1	0	
1	0	1	0	1	
0	1	1	0	1	
0	0	0	0	1	

Use truth tables of AND, OR, and NOT

Truth table for $(p \vee q) \wedge \neg(p \wedge q)$

p	q	$p \vee q$	$p \wedge q$	$\neg(p \wedge q)$	$(p \vee q) \wedge \neg(p \wedge q)$
1	1	1	1	0	0
1	0	1	0	1	1
0	1	1	0	1	1
0	0	0	0	1	0

Use truth tables of AND, OR, and NOT

Truth tables for complex formulas

- Intermediate columns: build complex expression step by step
- Each intermediate column is a **basic connective** (NOT,AND,OR,...) applied to already calculated previous columns
- Use truth tables of basic connectives to compute the value of new column, one at a time

4. Proving equivalence of two boolean expressions using truth tables

Which pair of expressions are equivalent to each other?

p	q	p&&q	!(p&&q)	!p	!q	!p !q
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

A. p, !p

B. p && q, !p || !q

C. q, !p

D. !(p&&q), !p || !q

Which pair of expressions are equivalent to each other?

p	q	p && q	!(p && q)	!p	!q	!p !q
1	1	1	0	0	0	0
1	0	0	1	0	1	1
0	1	0	1	1	0	1
0	0	0	1	1	1	1

A. p, !p

B. p && q, !p | !q

C. q, !p

D. !(p && q), !p | !q

Which pair of expressions are equivalent to each other?

p	q	p&&q	!(p&&q)	!p	!q	!p !q
1	1	1	0	0	0	0
1	0	0	1	0	1	1
0	1	0	1	1	0	1
0	0	0	1	1	1	1

So $!(p \& \& q) == !p \mid \mid !q$

Truth Table Example 1

a	b	c	!b	!b c	a && (!b c)
1	1	1	0	1	
1	1	0	0	0	
1	0	1	1	1	
1	0	0	1	1	
0	1	1	0	1	
0	1	0	0	0	
0	0	1	1	1	
0	0	0	1	1	

Truth Table Example 1

a	b	c	!b	!b c	a && (!b c)
1	1	1	0	1	
1	1	0	0	0	
1	0	1	1	1	
1	0	0	1	1	
0	1	1	0	1	0
0	1	0	0	0	0
0	0	1	1	1	0
0	0	0	1	1	0

Truth Table Example 1

a	b	c	!b	!b c	a && (!b c)
1	1	1	0	1	
1	1	0	0	0	0
1	0	1	1	1	
1	0	0	1	1	
0	1	1	0	1	0
0	1	0	0	0	0
0	0	1	1	1	0
0	0	0	1	1	0

Truth Table Example 1

a	b	c	!b	!b c	a && (!b c)
1	1	1	0	1	1
1	1	0	0	0	0
1	0	1	1	1	1
1	0	0	1	1	1
0	1	1	0	1	0
0	1	0	0	0	0
0	0	1	1	1	0
0	0	0	1	1	0

Truth Table Example 1

a	b	c	(a && b) !c
1	1	1	
1	1	0	
1	0	1	
1	0	0	
0	1	1	
0	1	0	
0	0	1	
0	0	0	

Truth Table Example 1

a	b	c	(a && b) !c
1	1	1	1
1	1	0	1
1	0	1	
1	0	0	
0	1	1	
0	1	0	
0	0	1	
0	0	0	

Truth Table Example 1

a	b	c	(a && b) !c
1	1	1	1
1	1	0	1
1	0	1	
1	0	0	1
0	1	1	
0	1	0	1
0	0	1	
0	0	0	1

Truth Table Example 1

a	b	c	(a && b) !c
1	1	1	1
1	1	0	1
1	0	1	0
1	0	0	1
0	1	1	0
0	1	0	1
0	0	1	0
0	0	0	1

Truth Table Example 1

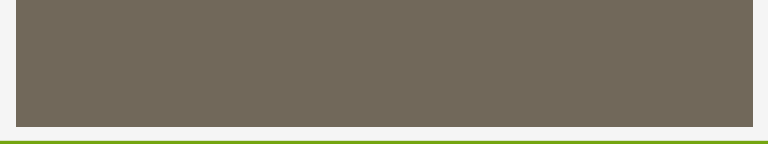
a	b	c	a && b	!c	(a && b) !c
1	1	1	1	0	1
1	1	0	1	1	1
1	0	1	0	0	0
1	0	0	0	1	1
0	1	1	0	0	0
0	1	0	0	1	1
0	0	1	0	0	0
0	0	0	0	1	1

How can we prove that two propositions are *not* equivalent?

- A. Make a truth table that has columns for both, and verify that not all the entries in the two columns are T
- B. Make a truth table that has columns for both, and verify that at least one of the entries in the two columns is F
- C. Make a truth table that has columns for both, and make sure that the two columns are not identical to each other.

How can we prove that two propositions are *not* equivalent?

- A. Make a truth table that has columns for both, and verify that not all the entries in the two columns are T
- B. Make a truth table that has columns for both, and verify that at least one of the entries in the two columns is F
- C. Make a truth table that has columns for both, and make sure that the two columns **are not identical to each other.**



Truth table for $(p \rightarrow q) \wedge \neg p$

p	q	$p \rightarrow q$	$\neg p$	$(p \rightarrow q) \wedge \neg p$
T	T	T	F	F
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

$(p \rightarrow q) \wedge \neg p$ is equivalent to:

- A. q
- B. $p \rightarrow q$
- C. $\neg p \vee q$
- D. $\neg p$
- E. Other/none/more than one